

Stationary models III

Version 1 (March 31, 2026)

ECON 31730, Spring 2026

University of Chicago

Department of Economics

Mikkel Plagborg-Møller

Order selection, bias-variance trade-off

- One very frequent model selection problem in time series analysis is selecting the VAR lag order p (and/or q in VARMA models).
- As always in statistics, we face a **bias-variance trade-off**:
 - ▶ Large p : low bias, high variance (many parameters to estimate).
 - ▶ Small p : potentially high bias, but low variance (few parameters).
- There exist many methods for selecting p based on data. We will see that they solve different problems. All of them cause issues for subsequent inference.
- We focus on VARs here for concreteness, but similar issues arise throughout econometrics. See Claeskens & Hjort (2008) for a very readable review.

Sequential testing

- Most obvious model selection strategy (“general to specific”):
 - ▶ Start with very large model $p = p_{\max}$.
 - ▶ Test $H_0: p = p_{\max} - 1$ against $H_1: p = p_{\max}$ (i.e., $H_0: A_{p_{\max}} = 0_{n \times n}$).
 - ▶ If don't reject, continue sequentially testing smaller models.
- Likelihood ratio test statistic for $H_0: p = \tilde{p} - 1$ vs. $H_1: p = \tilde{p}$:

$$2\widehat{LR}_{\tilde{p}-1|\tilde{p}} \equiv T \left\{ \log \det(\hat{\Sigma}^{(\tilde{p}-1)}) - \log \det(\hat{\Sigma}^{(\tilde{p})}) \right\},$$

with asy. distr. $\chi^2_{n^2}$ under H_0 (if u_t is i.i.d.). Here $\hat{\Sigma}^{(\tilde{p})}$ is the sample residual var-cov matrix for estimated VAR($\tilde{p}$).

- **Exercise:** Verify that for Gaussian VARs,

$$p(y_1, \dots, y_T \mid y_0, \dots, y_{1-p}, \hat{\theta}) = e^{-\frac{nT}{2}(1+\log(2\pi))} \det(\hat{\Sigma})^{-\frac{T}{2}}.$$

- Note: In these slides, T (not $T - p$) is the effective sample size.

Sequential testing (cont.)

- It's customary to use the same effective sample size T for estimating VARs of different orders p .
 - ▶ If $p_{\max}$ is the largest allowable lag order, we set first $p_{\max}$ observations aside as “pre-sample” observations.
 - ▶ Estimate $\text{VAR}(p)$ on T data points, using p “pre-sample” observations.
- Sequential testing has a positive probability of selecting a model that is too large, even as $T \rightarrow \infty$. Suppose we use significance level $\alpha \in (0, 1)$ (e.g., 5%) for the LR test. Then, for true $p < p_{\max}$,

$$\begin{aligned} P(\hat{p} > p) &\geq P(\text{reject } H_0: p = p_{\max} - 1) \\ &\rightarrow \alpha \quad \text{as } T \rightarrow \infty. \end{aligned}$$

- Could theoretically let $\alpha = \alpha_T \rightarrow 0$ as $T \rightarrow \infty$, but how to choose α_T in practice?

Bayesian model selection

- Bayes' rule: Given equal prior model probabilities, choose model M with highest **marginal likelihood** $p(y_1, \dots, y_T \mid M)$.
- We now derive a large- T approximation to the marginal likelihood.
- Define log likelihood fct $L_T(\theta_M \mid M) \equiv \log p(y_1, \dots, y_T \mid \theta_M, M)$ given model M . 2nd order Taylor expansion around MLE $\theta_M = \hat{\theta}_M$:

$$L_T(\theta_M \mid M) \approx L_T(\hat{\theta}_M \mid M) - \frac{1}{2}(\theta_M - \hat{\theta}_M)' \hat{H}_M(\theta_M - \hat{\theta}_M), \quad \hat{H}_M \equiv -\frac{\partial^2 L_T(\hat{\theta}_M \mid M)}{\partial \theta \partial \theta'}.$$

No first-order term, since $\hat{\theta}_M$ max'es likelihood.

- Insert into marginal likelihood:

$$\begin{aligned} p(y_1, \dots, y_T \mid M) &= \int \exp\{L_T(\theta_M \mid M)\} \pi(\theta_M \mid M) d\theta_M \\ &\approx \exp\{L_T(\hat{\theta}_M \mid M)\} \int \exp\{-\frac{1}{2}(\theta_M - \hat{\theta}_M)' \hat{H}_M(\theta_M - \hat{\theta}_M)\} \pi(\theta_M \mid M) d\theta_M \end{aligned}$$

Bayesian Information Criterion

- In the slide appendix, we use a **Laplace approximation** of the integral in the marginal likelihood to obtain

$$\log p(y_1, \dots, y_T \mid M) \approx$$

$$L_T(\hat{\theta}_M \mid M) - \frac{\dim(\theta_M)}{2} \log T + \log \pi(\hat{\theta} \mid M) + \frac{\dim(\theta_M)}{2} \log(2\pi) - \frac{1}{2} \log \det\left(\frac{1}{T} \hat{H}_M\right).$$

- For stationary models, LLN implies $\frac{1}{T} \hat{H}_M = O_p(1)$ as $T \rightarrow \infty$.
- The last three terms above are thus all asymptotically bounded as $T \rightarrow \infty$, while the first two terms diverge to infinity.
- Hence, for large T , choosing a model to max. the marginal likelihood is equivalent to minimizing the **Bayesian information criterion**

$$BIC(M) \equiv -\frac{2}{T} L_T(\hat{\theta}_M \mid M) + \dim(\theta_M) \frac{\log T}{T}.$$

Sometimes called the **Schwarz information criterion** (SIC).

- For VAR(p) models (neglecting a constant):

$$BIC(p) \equiv \log \det(\hat{\Sigma}^{(p)}) + n^2 p \frac{\log T}{T}.$$

Final Prediction Error

- Choosing p via BIC approximates Bayesian model selection.
- Different idea: Choose p to ensure good out-of-sample prediction.
- Use notation $\hat{\beta}$, y_t , and x_t like in previous slide deck. Assume i.i.d. innovations u_t .
- Now consider (slightly artificially) obtaining a new time series $\{\tilde{y}_t\}$ independent of – but from the same $\text{VAR}(p)$ model as – $\{y_t\}$. Let $\tilde{x}_t \equiv (1, \tilde{y}'_{t-1}, \dots, \tilde{y}'_{t-p})'$. Then $\tilde{y}_t = \beta \tilde{x}_t + \tilde{u}_t$, $\tilde{u}_t \stackrel{iid}{\sim} WN(0, \Sigma)$.
- Mean square forecast error for predicting $\tilde{y}_{T+1}$:

$$\begin{aligned}\text{Var}(\tilde{y}_{T+1} - \hat{\beta} \tilde{x}_T) &= \text{Var}(\tilde{u}_{T+1}) + E \left[(\hat{\beta} - \beta) \tilde{x}_T \tilde{x}_T' (\hat{\beta} - \beta)' \right] \\ &= \Sigma + E \left[(\hat{\beta} - \beta) S_x (\hat{\beta} - \beta)' \right],\end{aligned}$$

where $S_x \equiv E(x_t x_t')$. Here we used the law of iterated expectations and independence.

Final Prediction Error (cont.)

- The slide appendix uses the asymptotic variance of $\hat{\beta}$ to show that

$$\text{Var}(\tilde{y}_{T+1} - \hat{\beta}\tilde{x}_T) \approx \Sigma(1 + \frac{k}{T}),$$

where $k = 1 + np$ as usual, so $\beta \in \mathbb{R}^{n \times k}$.

- We would like to “minimize” the above forecast variance. But it’s a (positive definite) $n \times n$ matrix. One option is to minimize the log determinant of the matrix (sum of log eigenvalues):

$$\log \det(\text{Var}(\tilde{y}_{T+1} - \hat{\beta}\tilde{x}_T)) \approx \log \det(\Sigma) + n \log(1 + \frac{k}{T}).$$

- We also need to substitute an estimator of Σ . We use the degrees-of-freedom-adjusted estimator $\frac{T}{T-k} \hat{\Sigma}^{(p)} = \frac{1}{T-k} \sum_t \hat{u}_t^{(p)} (\hat{u}_t^{(p)})'$.
- Leads to the **Final Prediction Error** criterion:

$$FPE(p) \equiv \log \det(\hat{\Sigma}^{(p)}) + n \log \frac{T+1+np}{T-1-np}.$$

Akaike Information Criterion

- A related approach to model selection is to choose a model with high out-of-sample expected likelihood value at the MLE:

$$E_{\{\tilde{y}_t\}} \left[\log p(\tilde{y}_1, \dots, \tilde{y}_T \mid \hat{\theta}_M, M) \right],$$

where the MLE $\hat{\theta}_M$ is obtained from the original data set $\{y_t\}$. $\{\tilde{y}_t\}$ is a new data set, independent of the first one but with same distr.

- Corresponds to minimizing the **Kullback-Leibler divergence** between the estimated model and the true data generating process.
- Claeskens & Hjort (2008), ch. 2, show that we can approximate $(-\frac{2}{T} \times)$ the above objective by the **Akaike Information Criterion**

$$AIC(M) \equiv -\frac{2}{T} L_T(\hat{\theta}_M \mid M) + \dim(\theta_M) \frac{2}{T}.$$

- For VAR(p) models (neglecting constant):

$$AIC(p) \equiv \log \det(\hat{\Sigma}^{(p)}) + n^2 p \frac{2}{T}.$$

Summary of criteria

- BIC, FPE, and AIC are all of the form

$$IC(p) = \log \det(\hat{\Sigma}^{(p)}) + g(n, p, T).$$

Criterion	$g(n, p, T)$
BIC	$n^2 p \frac{\log T}{T}$
AIC	$n^2 p \frac{2}{T}$
FPE	$n \log \frac{T+1+np}{T-1-np}$

- Since $\log \frac{1+x}{1-x} \approx 2x$ for small x , FPE is large- T -equivalent w. AIC.
- BIC penalizes complex models more heavily than AIC/FPE.
 - ▶ BIC selects weakly smaller model than AIC if $T \geq \exp(2) \approx 7.4$.
- BIC+AIC used frequently in practice. Sometimes small-sample bias corrections are applied to the criteria, see Claeskens & Hjort (2008).

Consistency of information criteria

- An information criterion (IC) is said to be **consistent** if

$$\hat{p} \equiv \operatorname{argmin}_{\tilde{p} \leq p_{\max}} IC(\tilde{p}) \xrightarrow{P} p \quad \text{as } T \rightarrow \infty,$$

when the data is generated by a $\text{VAR}(p)$ with $p \leq p_{\max}$.

- Consider class of ICs (e.g., BIC, AIC, approximate FPE):

$$IC(p) \equiv \log \det(\hat{\Sigma}^{(p)}) + pg(T).$$

Proposition (Lütkepohl (2005), Prop. 4.2)

Let the true $\text{VAR}(p)$ be stationary with i.i.d. white noise innovations with finite 4th moments. Let $p \leq p_{\max}$. Then $IC(\cdot)$ is consistent if and only if $g(T) \rightarrow 0$ and $Tg(T) \rightarrow \infty$ as $T \rightarrow \infty$.

- BIC is consistent. AIC/FPE overestimate p with asy. prob. > 0 .

Efficiency of information criteria

- Consider the set-up in which we derived the FPE. Set $n = 1$. Define

$$MSFE(p) \equiv \text{Var}(\tilde{y}_{T+1} - \hat{\beta}_p \tilde{x}_T), \quad p = 0, 1, 2, \dots,$$

where $\hat{\beta}_p$ is the least-squares $\text{VAR}(p)$ estimator.

- Assume the data $\{\tilde{y}_t\}$ and $\{\tilde{x}_t\}$ follow a $\text{VAR}(\infty)$. Nevertheless, $MSFE(\cdot)$ could be minimized at some finite p , due to the bias-variance trade-off.
- We call an IC **efficient** if $\hat{p} = \underset{p \leq p_{\max}}{\text{argmin}} IC(p)$ satisfies

$$\frac{MSFE(\hat{p})}{\min_{\tilde{p} \leq p_{\max}} MSFE(\tilde{p})} \rightarrow 1 \quad \text{as } T \rightarrow \infty,$$

where we let $p_{\max} \rightarrow \infty$ at some appropriate rate.

- If true model is $\text{VAR}(\infty)$, AIC+FPE are efficient (as their derivation suggests). BIC is not. See Claeskens & Hjort (2008), ch. 4.5–4.6.

Conflict between consistency and efficiency

- Does any IC achieve both consistency and efficiency? Sadly, no.
- Consistency requires an IC to be very “conservative”: It must choose a small model whenever the data *might* be consistent with the small model. But then *worst-case* bias can be large, destroying efficiency.
- See Claeskens & Hjort (2008), ch. 4.9, for more discussion.
- We rarely think that the true model is precisely a $\text{VAR}(p)$ with finite p , so consistency does not appear that important for many applications.

Inference after model selection

- Model selection is almost always just the first step.
- Researchers often proceed with inference by pretending that the estimated $\hat{p}$ is the true p .
- If we're using a *consistent* IC, this is justified from the perspective of **pointwise asymptotics**: Given any true order p ,

$$P(\hat{\beta}_{\hat{p}} \in \mathcal{A}_T) = \underbrace{P(\hat{\beta}_p \in \mathcal{A}_T, \hat{p} = p)}_{\rightarrow P(\hat{\beta}_p \in \mathcal{A}_T) \text{ as } T \rightarrow \infty} + \underbrace{P(\hat{\beta}_{\hat{p}} \in \mathcal{A}_T, \hat{p} \neq p)}_{\rightarrow 0 \text{ as } T \rightarrow \infty},$$

for any set $\mathcal{A}_T$.

- Not necessarily comforting. It could still be that for *any* sample size T , there exists *some* true DGP such that the sampling distribution of $\hat{\beta}_{\hat{p}}$ is very different from that of $\hat{\beta}_p$:

$$\sup_{p, \beta_p} |P_{\beta_p}(\hat{\beta}_{\hat{p}} \in \mathcal{A}_T) - P_{\beta_p}(\hat{\beta}_p \in \mathcal{A}_T)| \not\rightarrow 0 \quad \text{as } T \rightarrow \infty.$$

“ $P_{\beta_p}(\cdot)$ ” = probability under a VAR(p) model w. param's β_p .

Inference after model selection (cont.)

for every p, β_p : $|P_{\beta_p}(\hat{\beta}_{\hat{p}} \in \mathcal{A}_T) - P_{\beta_p}(\hat{\beta}_p \in \mathcal{A}_T)| \rightarrow 0$ as $T \rightarrow \infty$,

$$\sup_{p, \beta_p} |P_{\beta_p}(\hat{\beta}_{\hat{p}} \in \mathcal{A}_T) - P_{\beta_p}(\hat{\beta}_p \in \mathcal{A}_T)| \not\rightarrow 0 \text{ as } T \rightarrow \infty.$$

- If an estimator $\hat{\beta}_{\hat{p}}$ satisfies the above two conditions, then we say it converges **non-uniformly**. Intuitively, the required sample size for the asymptotics to “kick in” may differ dramatically across parameters.
- Unfortunately, non-uniform convergence almost always occurs with inference after model selection.
- Hence, if we ignore the model selection step in our inference procedures, there usually exists *some* true model that will lead to poor performance, no matter how large is T .
- The issue is particularly severe for consistent ICs (e.g., BIC).
- Great reference: Leeb & Pötscher (2005).

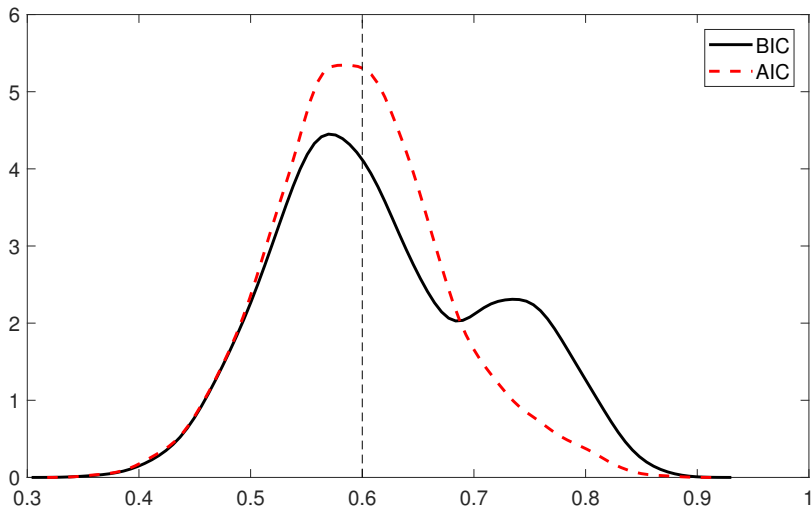
Inference after model selection: Simulation study

- Univariate AR(p) model:

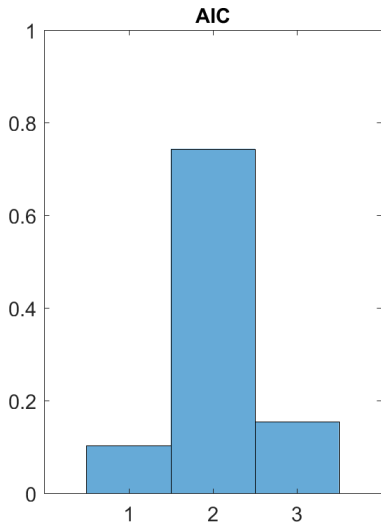
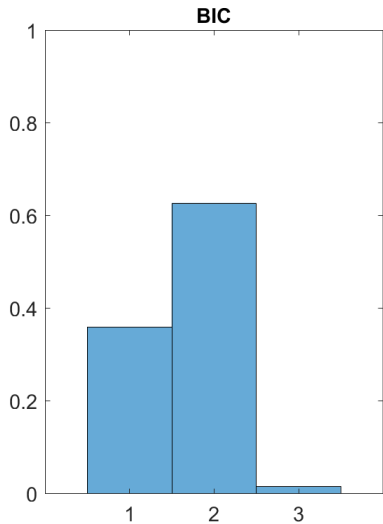
$$y_t = \nu + \sum_{\ell=1}^p \rho_{\ell} y_{t-\ell} + u_t, \quad u_t \stackrel{iid}{\sim} N(0, \sigma^2).$$

- True param's: $p = 2$, $\nu = 0$, $\rho_1 = 0.6$, $\rho_2 = 0.2$, $\sigma^2 = 1$ (stationary).
- Simulate fake data. Sample size $T = 200$.
- Estimate p by BIC/AIC, allowing $p \in \{1, 2, 3\}$. Estimate $\hat{\rho}_{\ell}$ ($1 \leq \ell \leq \hat{p}$) by least-squares given estimated lag order $\hat{p}$, with intercept.
- Repeat 10,000 times.

Simulation study: sampling density of $\hat{\rho}_1$



Simulation study: sampling distribution of $\hat{p}$



Inference after model selection (cont.)

- What should we do in practice?
- Some authors prefer fixing p *a priori*, i.e., avoiding data-dependent model selection altogether.
 - ▶ Typical choice: $p = 1, 4$, or 12 for annual, quarterly, or monthly data.
- Sensitivity to lag choice varies across inference methods as well as parameters of interest. More on this later when we discuss local projections.

Model averaging

- If all we want to do is forecast, perhaps we don't need to select a particular model (i.e., a single lag order p).
- We could instead *average* across results from different models, using weights that reflect the plausibility of each model.
- For example, **Bayesian model averaging**:

$$E(y_{T+1} \mid \text{data}) = \sum_{j=1}^J \underbrace{E(y_{T+1} \mid \text{data}, M_j)}_{\text{posterior mean given } M_j} \times \underbrace{P(M_j \mid \text{data})}_{\text{posterior prob. of } M_j} .$$

We know that the posterior model probabilities are tied to the marginal likelihood. Can be approximated by BIC.

- Other weighting schemes are also possible.

Appendix: Laplace approximation to marginal likelihood

$$p(y_1, \dots, y_T \mid M) \approx \exp\{L_T(\hat{\theta}_M \mid M)\} \int \exp\{-\frac{1}{2}(\theta_M - \hat{\theta}_M)' \hat{H}_M(\theta_M - \hat{\theta}_M)\} \pi(\theta_M \mid M) d\theta_M$$

- Change variables to $\vartheta = \sqrt{T}(\theta_M - \hat{\theta}_M)$. Jacobian of transformation:

$$\det\left(\frac{\partial \theta_M}{\partial \vartheta'}\right) = \det(T^{-1/2} I_{\dim(\theta_M)}) = T^{-\dim(\theta_M)/2}.$$

- We obtain the **Laplace approximation**

$$\begin{aligned} p(y_1, \dots, y_T \mid M) &\approx \exp\{L_T(\hat{\theta}_M \mid M)\} T^{-\dim(\theta_M)/2} \\ &\quad \times \int \exp\left\{-\frac{1}{2}\vartheta' \left(\frac{1}{T} \hat{H}_T\right) \vartheta\right\} \pi\left(\hat{\theta}_M + \frac{\vartheta}{\sqrt{T}} \mid M\right) d\vartheta \\ &\approx \exp\{L_T(\hat{\theta}_M \mid M)\} T^{-\dim(\theta_M)/2} \pi(\hat{\theta}_M \mid M) \\ &\quad \times \int \exp\left\{-\frac{1}{2}\vartheta' \left(\frac{1}{T} \hat{H}_M\right) \vartheta\right\} d\vartheta \\ &= \exp\{L_T(\hat{\theta}_M \mid M)\} T^{-\dim(\theta_M)/2} \pi(\hat{\theta}_M \mid M) \\ &\quad \times (2\pi)^{\dim(\theta_M)/2} \det\left(\frac{1}{T} \hat{H}_M\right)^{-1/2}, \end{aligned}$$

where the last line uses the usual formula for Gaussian integrals.

Appendix: Final Prediction Error derivation

$$\text{Var}(\tilde{y}_{T+1} - \hat{\beta}\tilde{x}_T) = \Sigma + E \left[(\hat{\beta} - \beta) S_x (\hat{\beta} - \beta)' \right]$$

- With i.i.d. innovations, we know that

$$\begin{aligned} \text{Var} \left(\text{vec} \left((\hat{\beta} - \beta) S_x^{1/2} \right) \right) &= (S_x^{1/2} \otimes I_n) \text{Var}(\text{vec}(\hat{\beta} - \beta)) (S_x^{1/2} \otimes I_n) \\ &\approx (S_x^{1/2} \otimes I_n) \left\{ \frac{1}{T} (S_x^{-1} \otimes \Sigma) \right\} (S_x^{1/2} \otimes I_n) = \frac{1}{T} (I_k \otimes \Sigma). \end{aligned}$$

Thus, each column $\hat{\xi}_j$ of $\hat{\xi} \equiv (\hat{\beta} - \beta) S_x^{1/2}$ has var-cov matrix $\frac{1}{T} \Sigma$.

- Hence,

$$E \left[(\hat{\beta} - \beta) S_x (\hat{\beta} - \beta)' \right] = E[\hat{\xi} \hat{\xi}'] = \sum_{j=1}^k E[\hat{\xi}_j \hat{\xi}_j'] = \frac{k}{T} \Sigma,$$

and

$$\text{Var}(\tilde{y}_{T+1} - \hat{\beta}\tilde{x}_T) \approx \Sigma \left(1 + \frac{k}{T} \right).$$